

PARKINSON DISEASE DETECTION USING VOICE SIGNAL WITH DEEP LEARNING

A PROJECT REPORT

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ABSTRACT

Parkinson's Disease is a progressive neurological disorder that primarily affects motor functions and speech, where early symptoms often manifest as subtle changes in voice characteristics such as reduced loudness, monotonic pitch, and vocal instability. Early diagnosis plays a crucial role in effective disease management; however, conventional diagnostic methods rely heavily on clinical expertise and specialized equipment, making them less accessible. To address this limitation, this work proposes a non-invasive and automated detection system based on deep learning techniques using voice signals. In the proposed approach, voice recordings are first preprocessed through normalization and noise reduction to enhance signal quality. Mel-Frequency Cepstral Coefficients (MFCC) are then extracted to represent essential speech features. A hybrid deep learning architecture combining Convolutional Neural Networks (CNN) and Bidirectional Long Short-Term Memory (BiLSTM) Autoencoder is employed. The CNN component captures spectral features, while the BiLSTM network models temporal dependencies in speech sequences. The autoencoder learns normal voice patterns and identifies abnormalities based on reconstruction error, which is compared against a predefined threshold to classify samples as healthy or affected. The proposed system offers a non-invasive, efficient, and cost-effective solution for early detection using simple voice recordings. It reduces reliance on manual diagnosis and facilitates timely medical intervention. With further optimization and the use of larger and more diverse datasets, the system has the potential to evolve into a reliable tool for real-world healthcare applications.

சுருக்கம்

பாரகி்ன்சன் ந ாய் என்பது உடல் இயக்கம் மற்றும் நபசச் ச பாதிக்கும் ஒரு முன்நனறும் ரம்பியல் குசறபாடு ஆகும். இ ஂத ந ாயின் ஆரம்ப அறிகுறிகள் பபரும்பாலும் குரல் மாற்றங்களாக காணப்படுகின்றன, உதாரணமாக குரல் சத்தம் குசறதல் மற்றும் pitch மாற்றங்கள். ஆரம்ப ிசலயில் ந ாசய கண்டறிதல் மிகவும் முக்கியமானது. ஆனால், பாரம்பரிய கண்டறிதல் முசறகள் மருத்துவ ிபுணரக் ள் மற்றும் சிறப்பு கருவிகசள நதசவயாக பகாண்டுள்ளதால் அசவ எளிதில் கிசடக்காது.இதசன தீரக் ஂக, இ ஂத திட்டத்தில் குரல் சிக்னல் கசள பயன்படுத்தி ஒரு தானியங்கி மற்றும் non-invasive கண்டறிதல் முசற முன்பமாழியப்படுகிறது. முதலில், குரல் பதிவுகள் சுத்திகரிக்கப்பட்டு (normalization மற்றும் noise removal) தரம் நமம்படுத்தப்படுகின்றன. பின்னர், Mel-Frequency Cepstral Coefficients (MFCC) அ ம் ச ங் க ள் எடுக்கப்படுகின்றன.இ ஂத அம்சங்கசள பயன்படுத்தி CNN மற்றும் BiLSTM Autoencoder இ ச ண ஂத deep learning மாதிரி பயன்படுத்தப்படுகிறது. இதில் CNN குரல் அம்சங்கசள கண்டறிகிறது மற்றும் BiLSTM கால வரிசச தகவல் கசள கற்றுக் பகாள் கிறது. Autoencoder மூலம் reconstruction error கணக்கிடப்பட்டு, அது ஒரு threshold உடன் ஒப்பிட்டு ந ாசய கண்டறிகிறது.இ ஂத முசற எளிய குரல் பதிவுகள் மூலம் விசரவாகவும் குசற ஂத பசலவில் ஆரம்ப ிசலயில் பாரகி்ன்சன் ந ாசய கண்டறிய உதவுகிறது. இது மருத்துவரக் ளின் நமல் சாரச் ப குசறத்து, ந ரச் மயான சிகிசச் சக்கு உதவுகிறது.

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LIST OF ABBREVIATIONS

CNN	Convolutional Neural Network
BiLSTM	Bidirectional Long Short-Term Memory
HNR	Harmonics-to-Noise Ratio
WAV	Waveform Audio File Format
MSE	Mean Squared Error
MFCC	Mel Frequency Cepstral Coefficients
ZCR	Zero Crossing Rate
RPDE	Recurrence Period Density Entropy
PPE	Pitch Period Entropy
GAN	Generative Adversarial Network
MP3	MPEG Audio Layer 3
API	Application Programming Interface
UI	User Interface
REST	Representational State Transfer
JSON	JavaScript Object Notation
SQL	Structured Query Language
CPU	Central Processing Unit
GPU	Graphics Processing Unit

CHAPTER 1

INTRODUCTION

Parkinson's Disease (PD) is a progressive neurodegenerative disorder that affects the movement and coordination of individuals. It is primarily caused by the loss of dopamine-producing neurons in the brain, leading to symptoms such as tremors, rigidity, slowed movement (bradykinesia), and speech impairments. As the disease progresses, it significantly impacts the quality of life of affected individuals.

Early detection of Parkinson's Disease is crucial for better management and treatment. However, traditional diagnostic methods are mostly based on clinical observation and may not identify the disease at an early stage. Recent studies have shown that changes in speech and voice patterns can act as early indicators of Parkinson's Disease.

This project focuses on developing an intelligent system for the early detection of Parkinson's Disease using voice signal analysis and machine learning techniques. The system processes voice recordings, extracts important features such as jitter, shimmer, and MFCC, and uses advanced models like CNN and BiLSTM Autoencoders to classify whether a person is healthy or affected. By providing a non-invasive, cost-effective, and automated solution, this project aims to assist in early diagnosis and support healthcare professionals in making accurate decisions.

1.1 Background

Voice-based analysis has become an effective approach for detecting neurological disorders, as changes in speech patterns can reflect underlying conditions. Traditional diagnostic methods are often costly and require clinical expertise, making early detection challenging. This project focuses on using deep learning techniques to analyze voice signals for identifying abnormalities. The system aims to provide a non-invasive, cost-effective, and accessible solution for early screening.

1.1.1 The context of the problem

The context of this project is the growing need for early detection of Parkinson's Disease (PD) using non-invasive and cost-effective methods. Parkinson's Disease is a progressive neurodegenerative disorder that affects movement and is commonly characterized by symptoms such as tremors, rigidity, and bradykinesia. Traditionally, diagnosis relies on clinical observation, which may delay early identification.

In recent years, research has shown that voice impairments (such as changes in pitch, tone, and stability) can serve as early indicators of Parkinson's Disease. This project focuses on leveraging voice signal analysis and machine learning techniques to detect these subtle changes.

The system aims to process voice recordings, extract relevant features (such as jitter, shimmer, and MFCC), and classify whether a person is healthy or shows signs of Parkinson's Disease. By using advanced models like deep learning and autoencoders, the project seeks to provide an automated, accurate, and early diagnostic support system.

This approach is significant because it enables:

- Early diagnosis and intervention
- Non-invasive and accessible screening
- Support for medical professionals in decision-making

Overall, the project lies at the intersection of healthcare and artificial intelligence, addressing the increasing prevalence of neurodegenerative disorders and the need for smarter diagnostic tools.

1.1 Problem Statement

Parkinson's Disease (PD) is a progressive neurological disorder that significantly affects movement, speech, and motor control. Early detection is essential for effective treatment and management, but current diagnostic methods are often expensive, time-consuming, and dependent on clinical expertise. Subtle voice impairments appear in the early stages of the disease, yet they are not effectively utilized for timely diagnosis.

There is a need for an automated, accurate, and non-invasive system that can analyze voice signals to detect early signs of Parkinson's Disease. Such a system should be capable of processing real-world voice data, identifying hidden patterns, and providing reliable predictions to assist healthcare professionals in early intervention.

Limitations of Existing Systems

Dependence on Traditional Machine Learning Models

Existing systems mainly use algorithms like SVM, KNN, and Decision Trees, which rely heavily on manual feature extraction and may not capture complex patterns effectively.

Limited Handling of Real-World Data

These models perform well in controlled environments but struggle with noisy data, variations in recording conditions, and diverse speech patterns.

Inability to Capture Temporal and Spectral Features Together

Voice signals are dynamic and sequential, but traditional models fail to effectively analyze both time-based and frequency-based characteristics.

Lack of Real-Time Implementation

Many systems are not designed for real-time usage, making them unsuitable for continuous monitoring or practical deployment.

Absence of Integrated Platforms

Existing solutions do not provide complete systems that include data processing, prediction, visualization, and report generation in one place.

Poor User Accessibility

Most systems are not user-friendly or web-based, limiting accessibility for common users and healthcare professionals.

No Automated Risk Assessment or Reporting

Lack of clear result interpretation, confidence scoring, and automated report generation reduces usability in real-world scenarios.

1.2 Objectives

The primary objective of this project is to develop an AI-based system for early detection of Parkinson's Disease using voice signals by analyzing speech patterns through deep learning techniques, and to provide an automated, non-invasive, and accessible diagnostic support system with result visualization and report generation.

The first objective is to develop an audio processing module that allows users to upload voice recordings through a web-based interface and supports multiple audio formats, including conversion of WebM files into WAV format using FFmpeg and PyDub for further analysis.

The second objective is to implement an audio preprocessing pipeline using the Librosa library, which includes resampling of audio signals to a standard frequency, normalization of amplitude, and removal of silence and noise to improve the quality and consistency of input data.

The third objective is to extract relevant acoustic features from speech signals using Mel-Frequency Cepstral Coefficients (MFCC), along with additional features such as Zero Crossing Rate (ZCR) and energy, to represent the spectral and temporal characteristics of the voice.

The fourth objective is to design and implement a hybrid deep learning model combining Convolutional Neural Network (CNN) and Bidirectional Long Short-Term Memory (BiLSTM) Autoencoder, where CNN is used to learn spatial and frequency-based patterns and BiLSTM captures sequential and temporal variations in the speech signal.

The fifth objective is to detect abnormalities in voice signals using reconstruction error (Mean Squared Error) generated by the autoencoder, and to classify the input as normal or Parkinson's affected based on a predefined threshold value.

Another important objective is to implement a risk assessment mechanism that categorizes the severity level (Low, Moderate, High) based on the deviation of reconstruction error from the threshold, providing better interpretability of results.

The next objective is to develop a database management system to store user details, uploaded audio files, prediction results, and history for future reference and analysis. The project also aims to implement a PDF report generation module using ReportLab, which includes patient details, prediction results, extracted features, graphical summaries, and medical recommendations.

Finally, the project aims to build a complete web-based system using Flask as the backend framework, integrating all modules such as user authentication, audio processing, prediction, result visualization, history tracking, and report generation, thereby providing a user-friendly and accessible platform for real-time Parkinson's Disease detection .

1.3 Scope of the Project

The proposed system for Parkinson's Disease detection leverages advanced deep learning techniques and audio signal processing to accurately analyze voice patterns and identify potential neurological abnormalities. The system utilizes speech signals as the primary input modality, providing a non-invasive and cost-effective approach for early diagnosis.

Voice recordings are collected from users through a web-based interface and are preprocessed using audio processing techniques such as normalization, silence removal, and resampling to ensure consistency and quality of the input data.

Feature extraction is performed using Mel-Frequency Cepstral Coefficients (MFCC), which effectively capture the spectral and perceptual characteristics of speech signals. In addition to MFCC, basic acoustic features such as Zero Crossing Rate (ZCR)

and signal energy are also computed to enhance the representation of voice patterns. These features are then fed into a hybrid deep learning model consisting of a Convolutional Neural Network (CNN) and a Bidirectional Long Short-Term Memory (BiLSTM) Autoencoder.

The CNN component is responsible for learning spatial and frequency-based features from the input data, while the BiLSTM layer captures temporal dependencies and sequential variations in speech.

The system detects abnormalities in voice signals by calculating the reconstruction error, specifically the Mean Squared Error (MSE), between the original input features and the reconstructed output generated by the autoencoder.

A predefined threshold value is used to classify the voice sample as either normal or Parkinson's affected. Furthermore, a risk assessment mechanism is incorporated to categorize the severity level into Low, Moderate, or High based on the deviation of the reconstruction error from the threshold, providing a more interpretable output for users.

The system is designed to operate efficiently on standard computing devices using commonly available resources such as microphones and web browsers, eliminating the need for specialized medical equipment. This ensures that the solution is scalable, accessible, and suitable for real-world deployment, particularly in remote and resource-limited environments.

By combining artificial intelligence with user-friendly web technologies, the proposed system offers a practical and effective solution for early screening and monitoring of Parkinson's Disease.

CHAPTER 2

LITERATURE SURVEY

2.1 INTRODUCTION

This chapter presents a comprehensive review of existing research on Parkinson's Disease detection, with specific focus on voice-based diagnostic approaches and deep learning methodologies. The reviewed papers span from foundational traditional machine learning methods to state-of-the-art deep learning architectures, providing the foundation and motivation for the proposed system.

2.2 REVIEW OF EXISTING SYSTEMS AND PAPERS

2.2.1 TITLE: CHARACTERIZING DISEASE PROGRESSION IN PARKINSON'S DISEASE FROM VIDEOS OF THE FINGER TAPPING TEST (2024)

AUTHOR(S): DIEGO L. GUARÍN, JOSHUA K. WONG, NIKOLAUS R. MCFARLAND, ADOLFO RAMIREZ-ZAMORA

This study developed an automated video-based assessment framework using markerless hand pose estimation via Google MediaPipe to analyze Finger Tapping test videos from the MDS-UPDRS III scale. The system extracted spatiotemporal features from hand joint positions and utilized machine learning classifiers to predict clinician-rated disease severity.

Limitations: The study relied on single clinician severity ratings, introducing potential subjectivity. The dataset was limited in size and suffered from class imbalance.

2.2.2 TITLE: EARLY DETECTION OF PARKINSON'S DISEASE USING DEEP NEUROENHANCENET(2024)

AUTHOR(S): TONGYUE HE, JUNXIN CHEN, XU XU, GIANCARLO FORTINO, WEI WANG

This work proposed Deep NeuroEnhanceNet, a CNN-based architecture incorporating Residual Blocks, Multi-head Attention mechanisms, and Squeeze-and-Excitation (SE) blocks for Parkinson's detection from smartphone walking recordings. The model demonstrated competitive accuracy for early-stage binary detection.

Limitations: The model performs only binary classification and cannot assess disease severity or progression.

2.2.3 TITLE: DETECTION OF EARLY PARKINSON'S DISEASE BY LEVERAGING SPEECH FOUNDATION MODELS (2025)

AUTHOR(S): QUANG DAO, LAETITIA JEANCOLAS, GRAZIELLA MANGONE ET AL.

This research investigated three large-scale speech foundation models — wav2vec2.0, Whisper, and SeamlessM4T — for PD detection using voice recordings. Transfer learning enabled high-accuracy classification using limited PD-specific labeled data.

Limitations: Dataset limited to early-stage PD. Gender imbalance affected model performance, particularly for female patients.

2.2.4 TITLE: MULTIMODAL DEEP LEARNING FRAMEWORK FOR PARKINSON'S DISEASE DETECTION (2025)

AUTHOR(S): MITESH KOTHARI, GANESHAN ARUMUGANAINAT, KALIRAJ KONAR

This study proposed a multimodal framework integrating multiple biomarker streams: voice signals analyzed using Random Forest, handwriting signals using XGBoost, gait analysis using ResNet-18, ground reaction force analysis using Transformer Encoders, and non-motor symptom analysis using Random Forest classifiers.

Limitations: Small dataset sizes for individual modalities increased overfitting risk. Clinical data collection settings limited real-world generalization.

2.2.5 TITLE: EARLY DIAGNOSIS OF PARKINSON'S DISEASE USING MACHINE LEARNING (2020)

AUTHOR(S): Z.K. SENTURK

This study applied SVM with feature selection for PD diagnosis from voice recordings. While effective on benchmark datasets, the approach required manual feature engineering and showed limited scalability with increasing dataset complexity.

Limitation: Dependence on handcrafted features limits generalization.

2.2.6 TITLE: INVESTIGATING VOICE AS A BIOMARKER FOR EARLY DETECTION OF PARKINSON'S DISEASE (2020)

AUTHOR(S): J.M. TRACY ET AL.

This work utilized Logistic Regression, Random Forest, and Gradient Boosting classifiers, demonstrating the predictive power of voice biomarkers. The study highlighted the need for multi-modal biomarker integration.

Limitation: Limited deep learning exploration; heavy reliance on handcrafted feature engineering.

2.2.7 TITLE: PARKINSON'S DISEASE DETECTION BASED ON SPECTROGRAM-DCGAN (2020),

AUTHOR(S): Z.J. XU ET AL.

This paper converted voice signals to spectrogram images and applied ResNet50 with GAN-based data augmentation to address class imbalance.

Limitation: Small training dataset; computational complexity of GAN training.

2.2.8 TITLE: RESNET-BASED PARKINSON'S DISEASE CLASSIFICATION (2023)

AUTHOR(S): OMAR EL ARISS, KAONING HU

This work applied ResNet50 with transfer learning and heat map conversion of TQWT, DWT, and MFCC features for classification. Limited to single-modality (speech) and the Sakar benchmark dataset.

Limitation: Class imbalance; restricted to single dataset evaluation. Future Work: Combine multiple feature sets; explore multimodal frameworks; investigate impact of color encoding

2.3 COMPARATIVE ANALYSIS

The following table presents a comparative overview of the reviewed methods, organized by methodology, features used, accuracy, and identified limitations:

SI.NO	TITLE	Author (Year)	Methodology and Features	Limitations	Future work
1	Characterizing Disease Progression in Parkinson's Disease from Videos of the Finger Tapping Test	Guarín et al. (2024)	MediaPipe + ML Spatiotemporal hand pose	Subjectivity, class imbalance	Future work includes collecting larger, balanced datasets, incorporating multiple clinician raters, and including de novo and late-stage PD patients.
2	Early Detection of Parkinson's Disease Using Deep NeuroEnhanceNet With Smartphone Walking Recordings	He et al. (2024)	NeuroEnhanceNet Gait (accelerometer)	Binary only, no severity grading	Future work includes extending analysis to severity assessment and symptom tracking, especially using rests tage data.
3	Detection of Early Parkinson's	Dao et al. (2025)	wav2vec2.0, Whisper Speech embeddings	Gender imbalance, early-stage	Future work includes adapting

	Disease by Leveraging Speech Foundation Models			only	models to noisier real world conditions (telephone and home recordings) using noise robust
4	Multimodal Deep Learning Framework for Customizabl e and Interpretable Parkinson's Disease Detection	Kothari et al. (2025)	Multimodal DL/ML Voice, handwriting, gait	Small modality datasets	Use of AlphaFoldbased protein structure prediction to link genotype with phenotype.
5	Early Diagnosis of Parkinson's Disease Using Machine Learning Algorithms	Senturk (2020)	SVM + Feature Selection Jitter, shimmer, HNR	Manual features, poor scalability	Automate feature extraction using deep learning

6	Investigating Voice as a Biomarker for Early Detection of Parkinson's Disease	Tracy et al. (2020)	Random Forest, LR Voice biomarkers	No deep learning	Integrate CNN-LSTM hybrid models; multimodal biomarker analysis
7	Parkinson's Disease Detection Based on Spectrogram-DCGAN	Xu et al. (2020)	ResNet50 + GAN Spectrograms	Small dataset, complex GAN training	Use larger datasets; optimize GAN stability; real-time implementation
8	ResNet-Based Parkinson's Disease Classification	El Ariss & Hu (2023)	ResNet50 Transfer Learning TQWT, DWT, MFCC	Single dataset, class imbalance	Combine multiple feature sets; explore multimodal frameworks; investigate impact of color encoding

Table 1 : Comparative Analysis

CHAPTER 3

PROPOSED SYSTEM

3.1 EXISTING SYSTEM

The existing approaches to Parkinson's Disease detection through voice analysis predominantly rely on classical machine learning techniques. The standard workflow involves three stages: manual feature extraction, classifier training, and binary classification.

3.1.1 Working of Existing Systems:

Voice recordings are collected from standard benchmark datasets such as the UCI Parkinson's Dataset. Acoustic features including jitter, shimmer, Harmonic-to-Noise Ratio (HNR), and Recurrence Period Density Entropy (RPDE) are manually extracted. Classifiers such as SVM, KNN, or Decision Tree are trained on extracted features. A binary label (Normal/PD) is predicted for new samples.

3.1.2 Drawbacks of Existing Systems:

- Manual feature engineering requires domain expertise and is time-consuming.
- Traditional classifiers cannot learn hierarchical representations or temporal patterns from sequential voice data.
- Systems perform poorly on noisy, real-world audio recordings outside controlled conditions.
- No risk stratification — outputs only binary classification without severity indication.
- Lack of end-to-end deployable web applications with report generation capabilities.
- Batch processing of multiple voice samples simultaneously is unsupported.

3.2 PROPOSED SYSTEM

The proposed system overcomes the limitations of traditional approaches by employing a deep learning-based end-to-end pipeline for Parkinson's Disease detection from voice signals. The system integrates MFCC-based feature extraction with a hybrid CNN + BiLSTM Autoencoder model and is deployed as a responsive Flask web application.

3.2.1 Overview of Proposed System:

Voice samples are accepted as uploaded WAV files, batch folder uploads, or real-time microphone recordings via a web browser. MFCC features are automatically extracted from preprocessed audio, eliminating manual feature engineering. A CNN sub-network learns local spectral patterns and frequency-domain representations. A BiLSTM sub-network processes the output sequentially to capture temporal and contextual voice dynamics. An Autoencoder architecture trains on healthy voice samples to learn the normal reconstruction pattern; PD samples exhibit higher reconstruction error (MSE). An MSE-based threshold determines classification: below threshold = Normal; above = Parkinson's Detected. Risk levels (Low, Moderate, High) are assigned based on MSE magnitude ranges.

3.2.2 Features of Proposed System:

- Fully automated feature extraction — no manual intervention required.
- Superior accuracy through deep temporal-spatial feature learning.
- Risk stratification provides actionable clinical information beyond binary diagnosis.
- Supports single-file, folder batch processing, and real-time recording.
- Generates downloadable PDF clinical reports with timestamps and patient identification.
- Provides visual MSE distribution graphs for session-level analysis.
- User authentication ensures secure, multi-user clinical deployment.

3.3 SYSTEM ARCHITECTURE

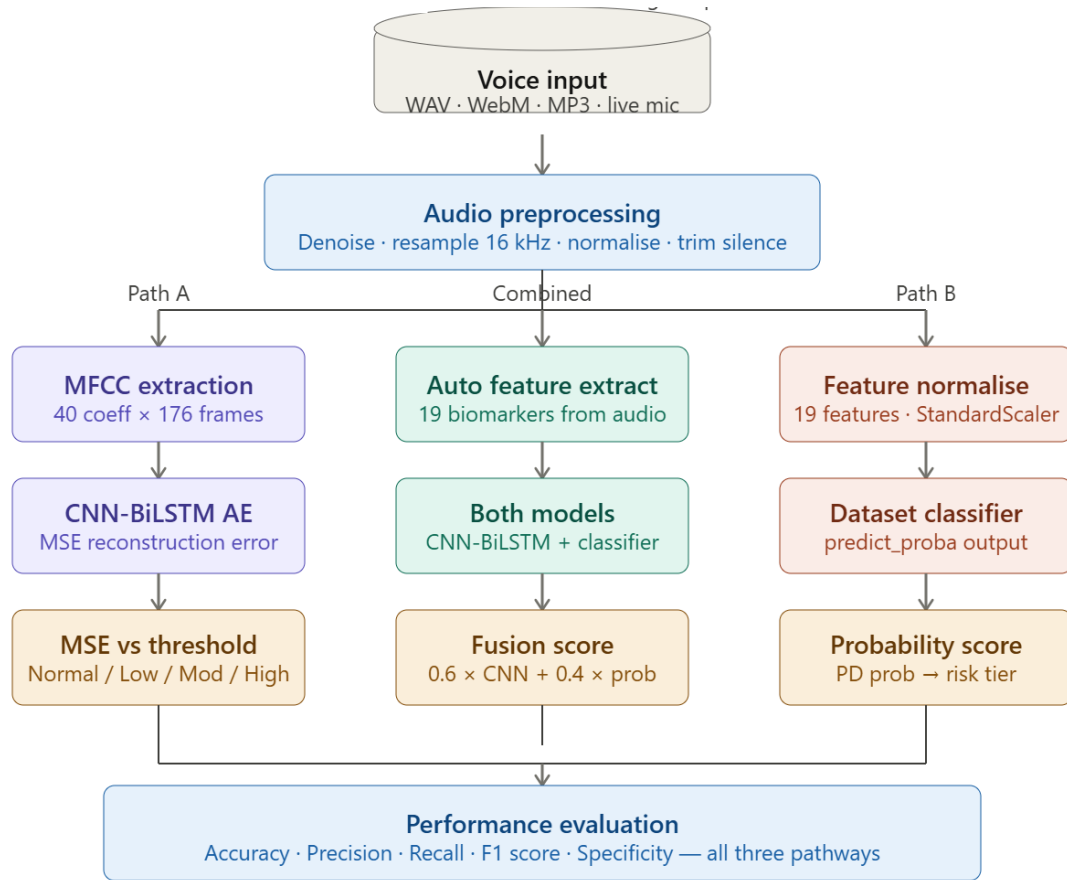


Figure 1: Block Diagram

The system architecture consists of five primary modules interconnected to form the end-to-end detection pipeline. Each module is described below:

Data Input Layer: Accepts voice inputs via file upload (WAV, MP3, WebM), folder-based batch upload, or real-time browser microphone recording using the Web Audio API.

Preprocessing and Feature Extraction: Audio signals are normalized to a consistent amplitude range, silence is removed, and MFCC features (40 coefficients) are

extracted using the Librosa library. Features are padded or truncated to a fixed length of 862 time steps to produce a uniform (862×40) input tensor.

CNN + BiLSTM Autoencoder Model: The encoder phase uses 1D Convolutional layers to detect local spectral patterns, followed by BiLSTM layers to capture bidirectional temporal context. The decoder mirrors the encoder with upsampling to reconstruct the input. The model is trained exclusively on healthy voice samples, minimizing Mean Squared Error (MSE) loss.

Classification Engine: Reconstruction MSE is computed for each input sample. Samples are classified as Normal ($\text{MSE} \leq \text{threshold}$) or Parkinson's Detected ($\text{MSE} > \text{threshold}$). Risk levels are assigned: Low Risk (MSE slightly above threshold), Moderate Risk ($1.5 \times \text{threshold}$), High Risk ($> 2 \times \text{threshold}$).

Output and Reporting: Results are displayed in a responsive web dashboard. A PDF clinical report is generated using ReportLab containing patient data, per-file MSE values, status badges, risk levels, and an overall assessment. Prediction history is stored in an SQLite database.

3.4 METHODOLOGY

The proposed system follows a systematic pipeline to analyze voice signals and classify them into normal and abnormal categories using deep learning techniques. The methodology consists of the following stages:

3.4.1 STEP BY STEP WORKING OF THE SYSTEM

The system takes a voice recording as input and preprocesses it by normalizing the signal, removing silence, and reducing noise. It then extracts features such as MFCC and converts them into a fixed-size format. The processed data is fed into a CNN-BiLSTM model to learn patterns and temporal characteristics. Finally, the system classifies the voice as normal or abnormal and displays the result.

Step 1: Data Collection

Voice recordings are collected from two categories: healthy individuals (normal class) and Parkinson's Disease (PD) patients. This dataset forms the foundation for training and evaluating the model.

Step 2:Audio Preprocessing

Each audio file is processed by loading the signal, normalizing amplitude for consistency, and removing silence segments to retain meaningful speech. Then, 40 MFCC features are extracted using a 25 ms window and 10 ms hop length to capture important speech characteristics.

Step 3:Feature Standardization

The extracted MFCC features are standardized by padding or truncating them to a fixed shape of (862, 40). This ensures uniform input size, which is essential for deep learning model training.



Figure 2 : Accoustic feature importance:mean values for PD vs Healthy voice

Step 4:Model Training (Autoencoder)

A hybrid CNN + BiLSTM autoencoder model is trained using only healthy voice data. The model learns normal speech patterns by capturing both spectral features (via CNN) and temporal dependencies (via BiLSTM).

Step 5: Model Optimization

The model is optimized using Mean Squared Error (MSE) as the loss function and the Adam optimizer. Training is performed over multiple epochs to minimize reconstruction error and improve model performance.

Step 6: Threshold Calculation

After training, the model is evaluated on validation data to compute the reconstruction error distribution. A threshold is set at the 95th percentile to distinguish between normal and abnormal voice samples.

Step 7: Model Saving

The trained model is saved in a .keras file, and the computed threshold value is stored in a .pkl file for use during prediction.

Prediction / Inference Phase

User input is provided through a web application either by uploading or recording voice. The same preprocessing steps are applied, and the trained model is loaded to process the input.

Reconstruction and Classification

The input is passed through the autoencoder to calculate reconstruction error (MSE). This error is compared with the predefined threshold to classify the voice as normal or Parkinson's detected.

Risk Assessment

Based on the reconstruction error, the system assigns a risk level such as low, moderate, or high, providing more detailed insights beyond simple classification.

Data Storage and Result Display

All prediction results are stored in a SQLite database. The system displays outcomes on a web interface, including prediction status, risk level, and summary analysis.

Report Generation

Finally, the system generates a downloadable PDF report containing input details, prediction results, and risk assessment for user reference.

3.4.2 FLOW CHART

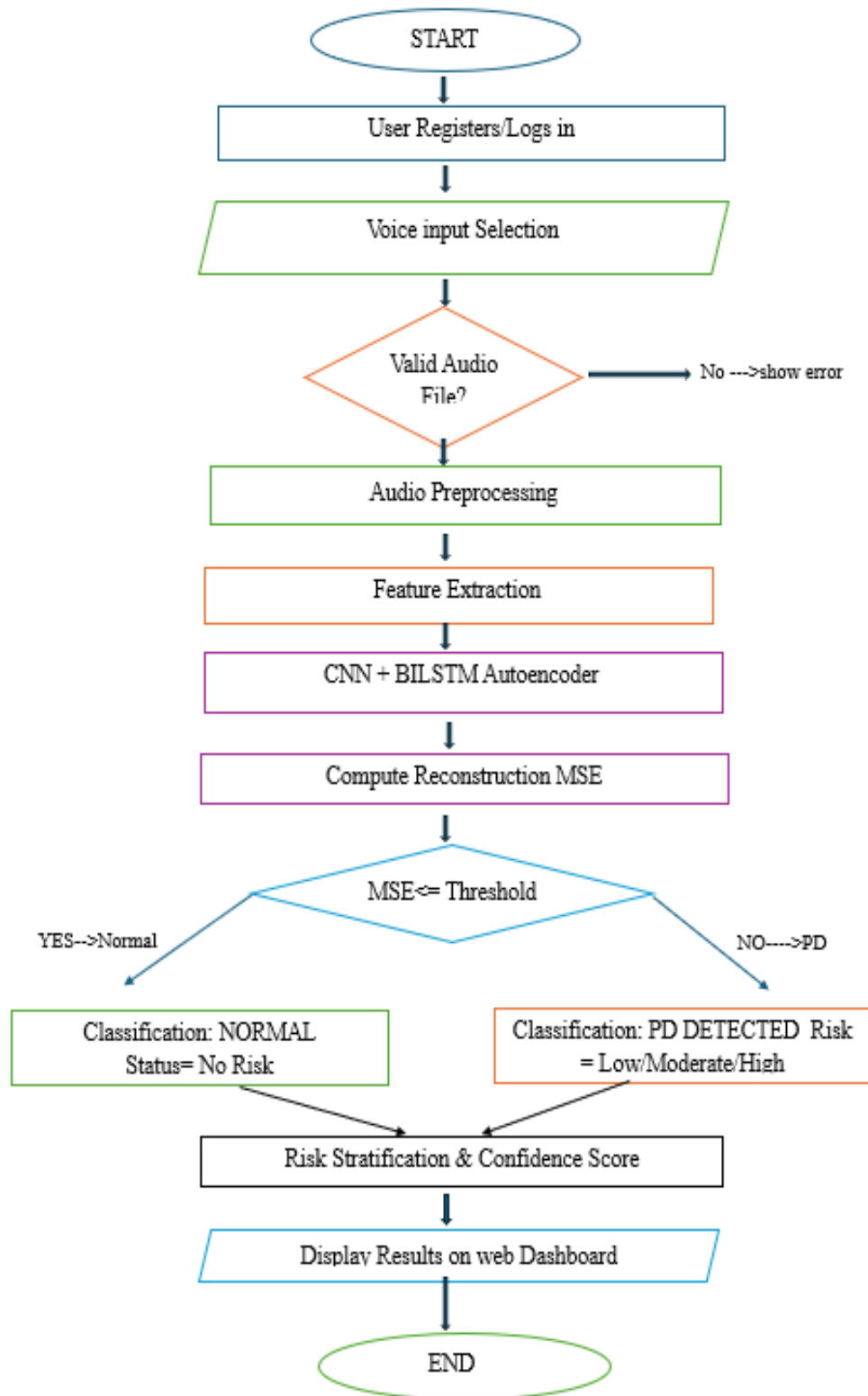


Figure 3: Flow chart

3.5 MODULE DESCRIPTION

3.5.1: USER AUTHENTICATION & SESSION MANAGEMENT

This module manages user registration and login using hashed password storage in an SQLite database. It ensures that each user's prediction history is isolated and securely maintained. Flask session tokens are used to maintain login state. Only authenticated users may access the diagnostic dashboard or prediction history.

3.5.2: VOICE INPUT & PREPROCESSING

This module handles three input modes: file upload (single or multiple WAV/audio files), folder directory upload (batch processing), and real-time browser recording using the MediaRecorder Web API. Audio preprocessing involves amplitude normalization using Librosa's load function, silence trimming, MFCC extraction (40 coefficients, sr=22050), and zero-padding/truncation to a standardized 862-timestep length. The resulting feature tensor of shape (862, 40) is fed into the deep learning model.

3.5.3: CNN + BILSTM AUTOENCODER

The core deep learning model is a CNN + BiLSTM Autoencoder. The encoder consists of two 1D Convolutional layers (filters: 64, 128) with ReLU activation and MaxPooling, followed by two BiLSTM layers (units: 64, 32) that process the feature sequence in both forward and backward temporal directions. The decoder reconstructs the input using RepeatVector, BiLSTM, and TimeDistributed Dense layers.

3.5.4: CLASSIFICATION & RISK STRATIFICATION

After computing per-sample MSE, the classification engine applies the stored threshold to label each voice as Normal or Parkinson's Detected. Risk stratification assigns Low Risk for MSE values slightly above the threshold (1.0x – 1.5x), Moderate Risk for values 1.5x – 2.0x, and High Risk for values exceeding 2.0x. An overall session risk level is computed based on the distribution of individual file risk levels.

3.5.5: WEB INTERFACE & REPORTING

The Flask-based web interface provides a responsive, professional dashboard accessible via any modern browser. Pages include: Login, Register, Dashboard (with statistics, upload, recording, and doughnut chart), Results Page (with per-file table and MSE graph), and History Page (with search/filter and export to Excel). PDF reports are generated using the ReportLab library containing patient session details, tabulated per-file results, and a clinical interpretation note.

CHAPTER 4

IMPLEMENTATION AND RESULTS

4.1 TOOLS AND TECHNOLOGIES USED

4.1.1 Software Requirements

Component	Specification
Operating System	Windows 10/11 or Linux (Ubuntu 20.04+)
Programming Language	Python 3.8 or higher
Development Environment	Jupyter Notebook / VS Code / PyCharm
Web Framework	Flask 2.x
Deep Learning	TensorFlow 2.x / Keras
Audio Processing	Librosa 0.9+
Data Handling	NumPy, Pandas
Visualization	Matplotlib, Chart.js
Model Persistence	Pickle (.pkl), Keras H5/.keras format
Report Generation	ReportLab
Database	SQLite3
Dataset	ADRS, AH (MDVoice), HC, PD benchmark WAV datasets

Table 2: Software Requirements

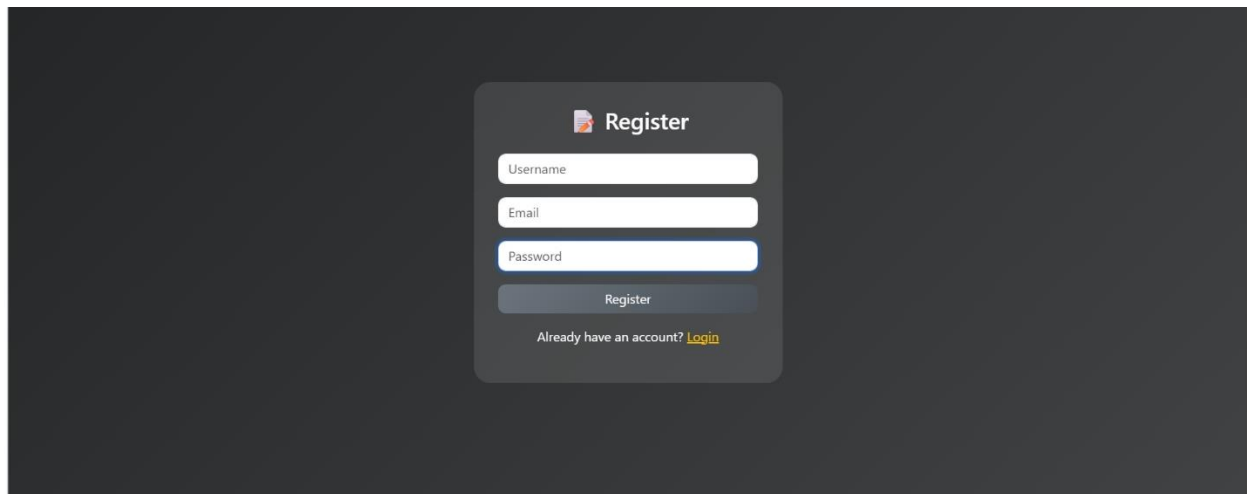
4.1.2 Hardware Requirements

Component	Specification
Processor	Intel i5 7th Gen or higher (GPU optional: NVIDIA GTX 1050+)
RAM	Minimum 8 GB (16 GB recommended for training)
Storage	10–20 GB free disk space
Microphone	Any standard input device (for live recording)
Browser	Google Chrome 90+ / Microsoft Edge / Firefox

Table 3 : Hardware Requirements

4.2 SYSTEM IMPLEMENTATION

4.2.1 Register page



The image shows a dark-themed user interface for a registration page. In the center, there is a light gray rounded rectangle containing the registration form. At the top of this rectangle is a document icon followed by the word "Register". Below this are three input fields: "Username", "Email", and "Password". The "Password" field has a blue border. Under the input fields is a dark gray button labeled "Register". At the bottom of the form, there is a link that says "Already have an account? [Login](#)".

Figure 4: Register page

The Registration Page allows a new user to create an account before accessing the Parkinson voice screening system. The user enters basic details such as username, email address, and password. After successful registration, the user can log in and use the voice analysis, numerical biomarker, symptom guidance, and history modules. This page helps maintain separate user records, so each user's voice analysis results and history are stored under their own account. It also improves system security by allowing only registered users to access the dashboard and prediction features.

4.2.2 Login Page

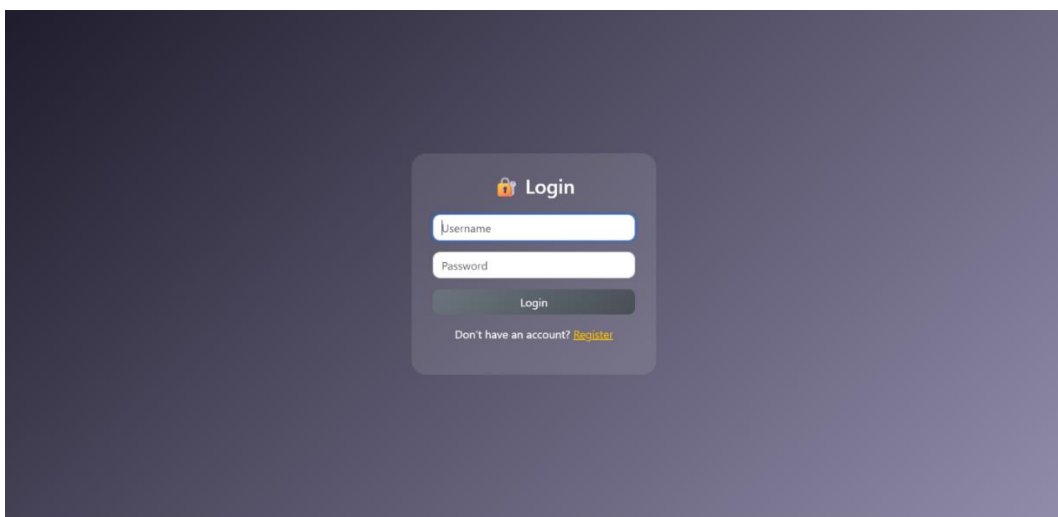


Figure 5: Login Page

The login page is the entry point of the Parkinson voice screening system. It allows only registered users to access the application by entering their username and password. This page helps maintain user privacy and ensures that each user's test results and history are stored under their own account. After successful login, the user is redirected to the clinical dashboard.

4.2.3 Clinical Dashboard

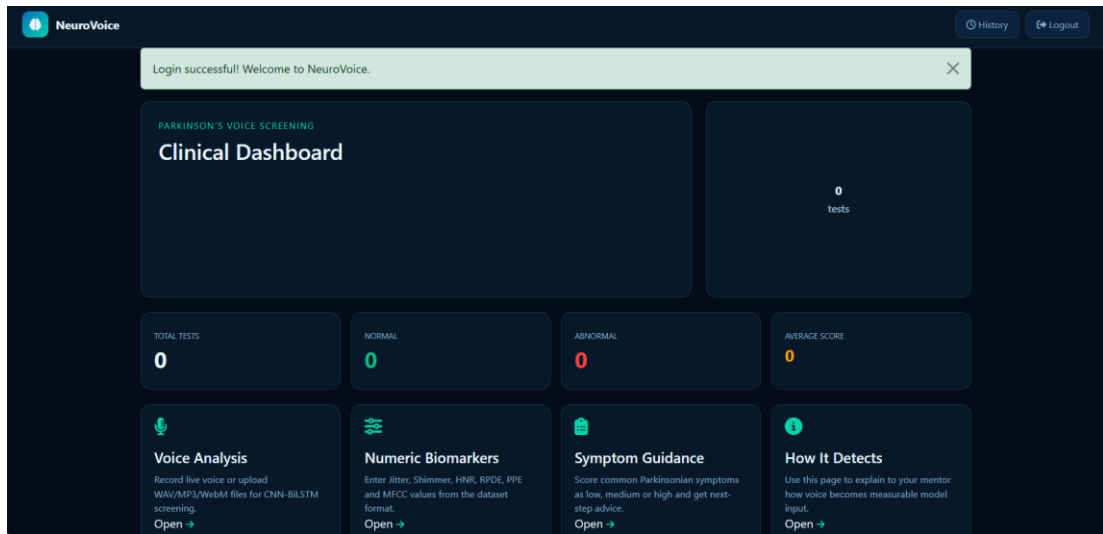


Figure 6: Clinical Dashboard

The clinical dashboard provides an overview of the entire system. It displays summary information such as the total number of tests conducted, number of normal results, number of abnormal results, and the average score. From this page, the user can easily navigate to voice analysis, numerical biomarkers, symptom guidance, model explanation, and history. This page acts as the main control center of the application.

4.2.4 Voice Analysis Page

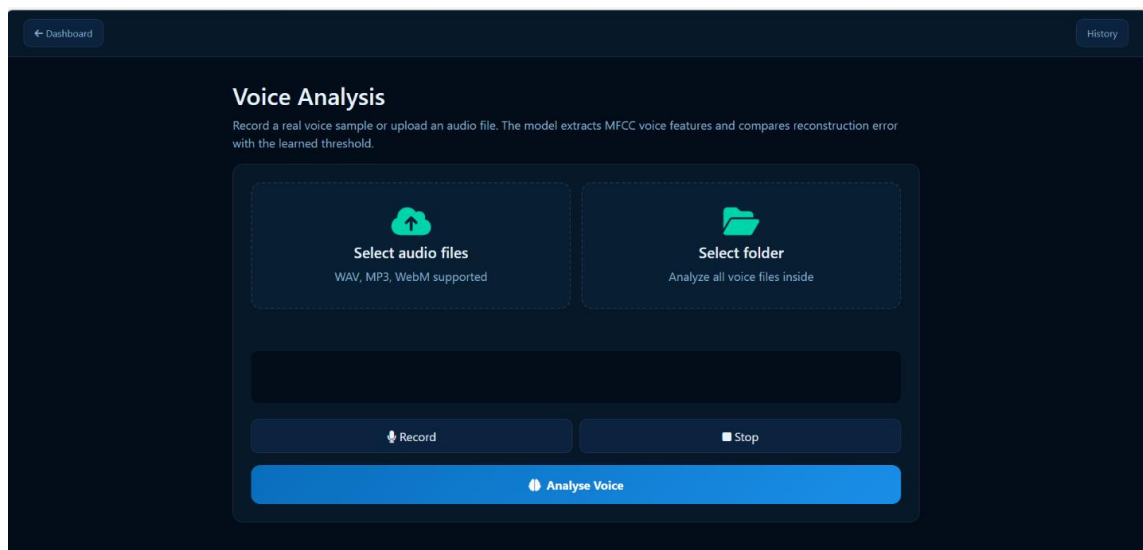


Figure 7: Voice Analysis Page

The voice analysis page is used to provide voice input for Parkinson screening. Users can either upload an audio file, select a folder containing multiple voice samples, or record a live voice sample. The uploaded voice is processed by the system, converted into acoustic features, and analyzed using the trained model. This page helps identify whether the voice pattern is normal or shows Parkinson-like characteristics.

4.2.5 Numeric Voice Biomarkers

← Dashboard History

Numeric Voice Biomarkers

Enter dataset-style values when you want to test known feature ranges instead of uploading audio.

Jitter	Shimmer	ZCR
HNR	RPDE	PPE
MFCC0	MFCC1	MFCC2
MFCC3	MFCC4	MFCC5
MFCC6	MFCC7	MFCC8
MFCC9	MFCC10	MFCC11
MFCC12		

Compare With Dataset

Figure 8: Numeric Voice Biomarkers

This page is used to enter numerical voice features instead of uploading an audio file. The user provides values such as Jitter, Shimmer, ZCR, HNR, RPDE, PPE, and MFCC. These features describe pitch variation, loudness variation, noise level, and voice frequency patterns. The system analyzes these values and predicts whether the sample is normal or Parkinson-like.

4.2.6 Symptoms Page

← Dashboard

How it detects

Symptom Severity Guidance

NO VOICE RESULT IN THIS SESSION
Use Voice Analysis first if you want the final result to combine voice signal and symptoms.

Resting tremor	0	1	2	3	4	0
Slowness of movement	0	1	2	3	4	0
Muscle stiffness	0	1	2	3	4	0
Balance difficulty	0	1	2	3	4	0
Soft or unclear voice	0	1	2	3	4	0
Small handwriting	0	1	2	3	4	0

Reduced smell	0	1	2	3	4	0
Sleep movement problems	0	1	2	3	4	0

0 = none, 1 = low, 2 = mild, 3 = moderate, 4 = high

Get Suggestion

Figure 9: Symptoms Page

The symptoms page collects additional symptom information from the user. It includes common Parkinson-related symptoms such as resting tremor, slowness of movement, muscle stiffness, balance difficulty, soft voice, small handwriting, reduced smell, and sleep movement problems. The user selects severity levels from 0 to 4. Based on the selected symptoms, the system provides supportive guidance and combines it with the voice analysis result for better screening support.

4.3 WORKING MODEL — STEP-BY-STEP EXECUTION

Step 1. User Login

The user first opens the NeuroVoice web application. The user logs in using registered username and password. After successful login, the user is redirected to the dashboard.

Step 2. Voice Input Selection

From the dashboard or voice analysis page, the user can provide voice input in three ways: upload a single audio file, select a folder containing multiple audio files, or record live voice using the browser microphone.

Step 3. Audio File Upload

After submitting the form, the selected or recorded audio files are sent to the Flask backend. The files are stored in the server's uploads folder for processing.

Step 4. Feature Extraction

The /predict route processes each uploaded file one by one. The system extracts important voice features such as MFCC, Jitter, Shimmer, HNR, PPE, and other acoustic biomarkers from the audio signal.

Step 5. CNN-BiLSTM Model Processing

The extracted MFCC features are passed into the pre-trained CNN-BiLSTM Autoencoder model. The model tries to reconstruct the input feature pattern.

Step 6. MSE Calculation

The system calculates Mean Squared Error between the original MFCC features and the reconstructed MFCC features. MSE shows how much difference exists between the input and reconstructed output.

Step 7. Prediction and Risk Classification

The MSE value is compared with the saved threshold value. If the MSE is below the threshold, the voice is classified as Normal. If the MSE is above the threshold, it is classified as Abnormal or Parkinson-like. Risk level is assigned as Low, Moderate, or High based on the severity.

Step 8. Database Storage

The prediction result is stored in the SQLite database. The stored details include filename, MSE value, prediction result, risk level, audio quality values, Parkinson probability, and timestamp.

Step 9. Result Display

After analysis, the user is redirected to the result page. The page displays the final

prediction, risk level, MSE score, threshold value, Parkinson probability, extracted biomarkers, and the reason for the result.

Step 10. Report and History

The user can download a PDF report, export results as CSV, or view previous predictions in the history page. This helps users keep track of earlier voice analysis records.

4.4 RESULTS AND OUTPUTS

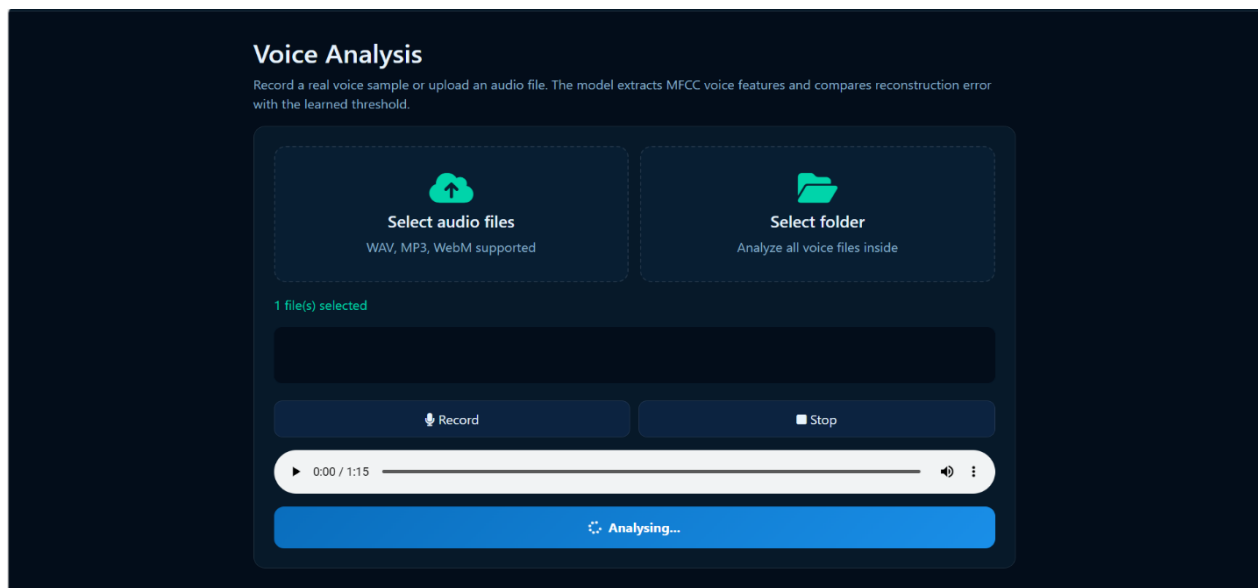


Figure 10: Raw voice Analysis Input

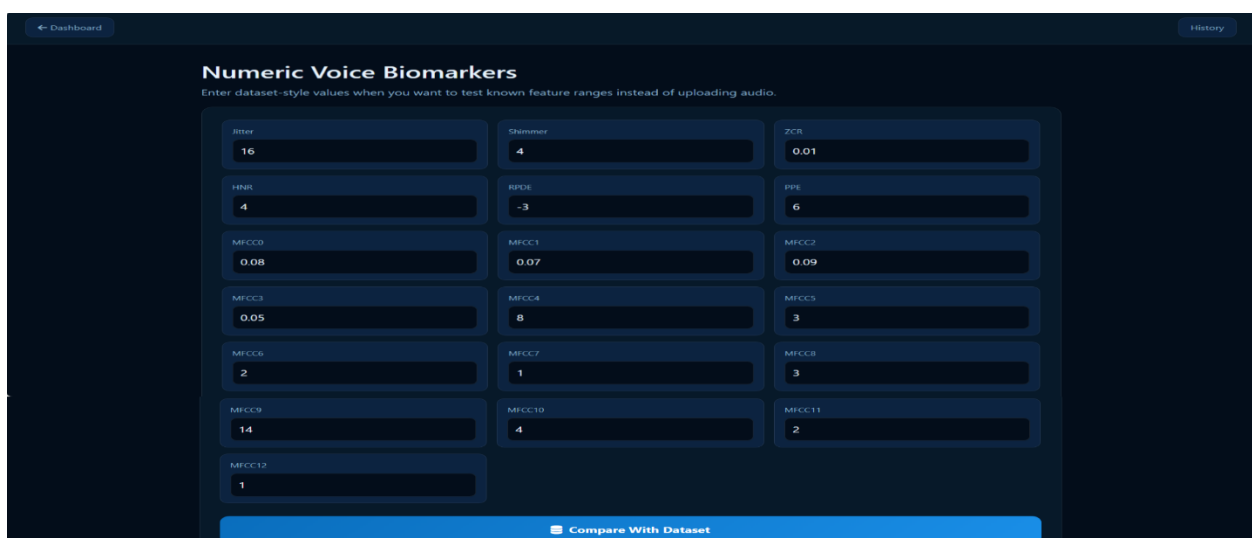
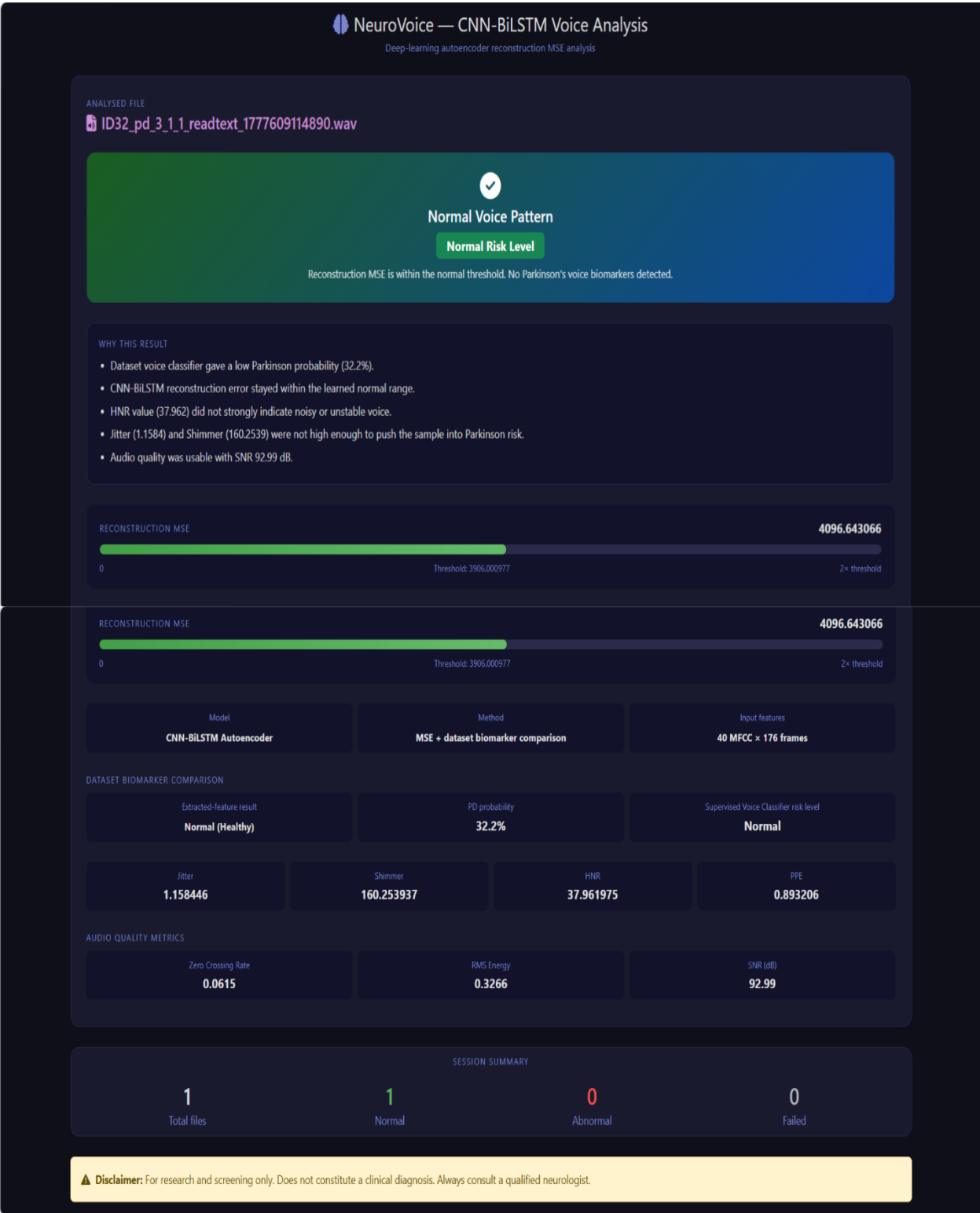


Figure 11: Numeric voice Input



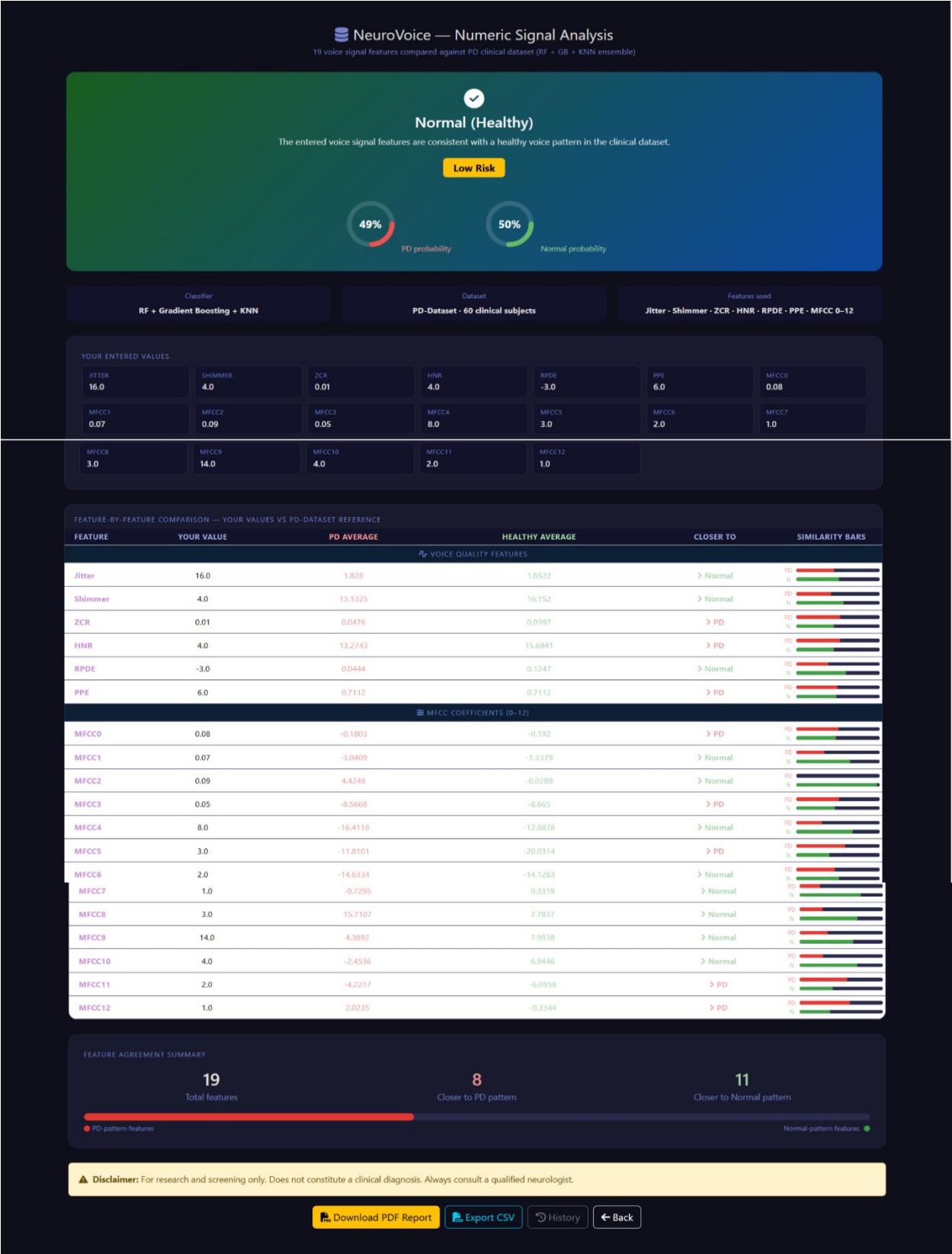


Figure 13 : Numeric voice analysis Result

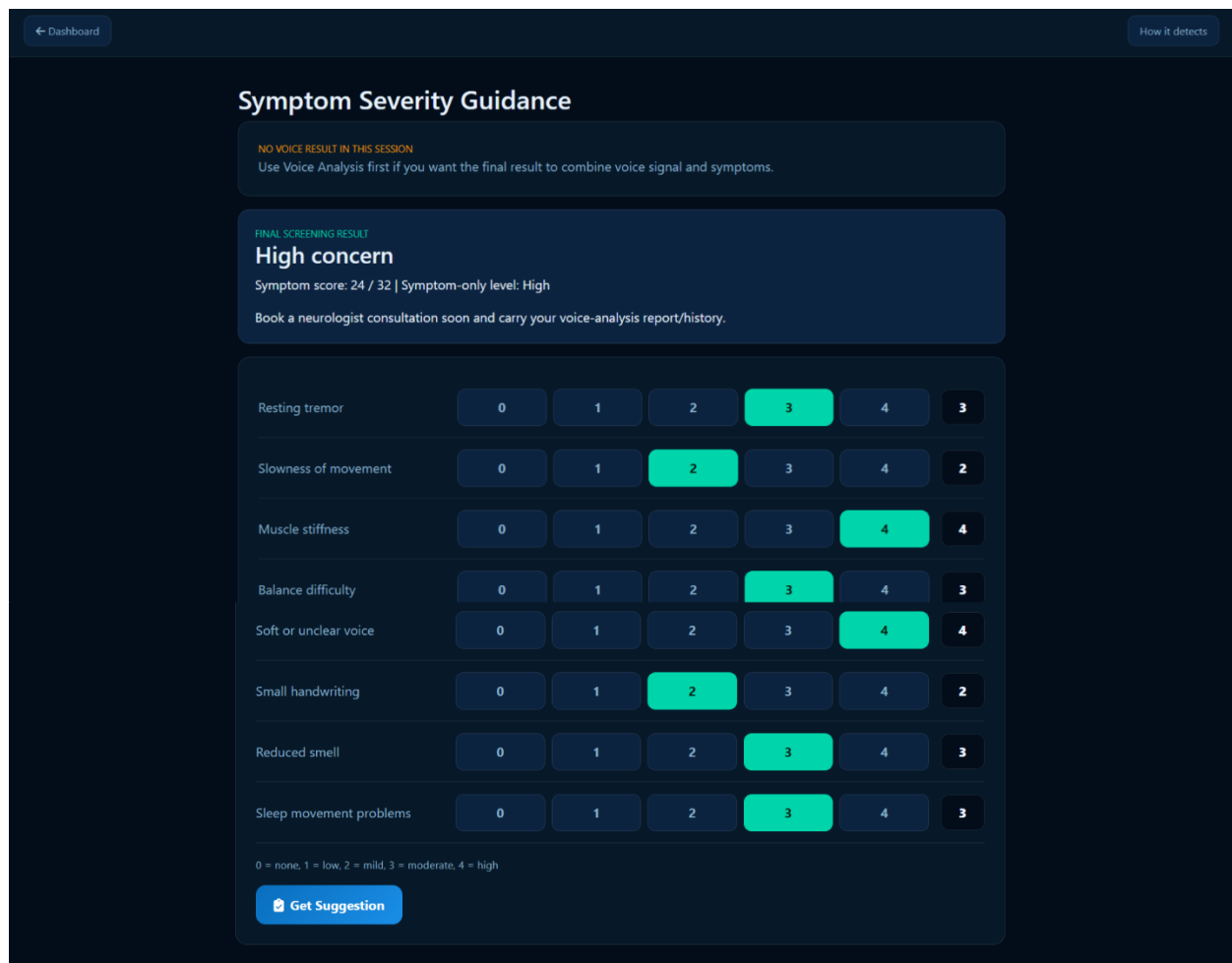


Figure 14 :Symptoms Severity Result

The system produces the following outputs upon processing voice samples:

Dashboard Statistics: The main dashboard displays three real-time counters Total Tests Conducted, Normal Cases, and Abnormal Cases — updated after each prediction session.

Results Table: Each submitted audio file appears as a row in the results table with columns: Filename, MSE Score, Status (Normal / Abnormal), and Risk Level (Low / Moderate / High Risk). Status and risk are displayed using color-coded badge pills.

Overall Risk Banner: An overall session risk classification (Normal, Moderate Risk, or High Risk) is displayed prominently at the top of the results page, accompanied by a clinical interpretation message.

MSE Graph: A Matplotlib-generated scatter/line plot displays the MSE reconstruction error for each sample in the session, with a horizontal threshold line for visual reference. Samples above the line are PD-suspected.

PDF Clinical Report: A downloadable PDF report is generated using ReportLab, containing the patient/session identifier, analysis timestamp, per-file tabulated results, and a clinical interpretation summary. The report is formatted for direct use in clinical records.

4.5 PERFORMANCE ANALYSIS

The system’s performance is evaluated using metrics such as accuracy, precision, recall, and F1-score to ensure reliable classification of voice signals. The model demonstrates effective learning of speech patterns, achieving high accuracy in distinguishing normal and abnormal cases. Results indicate that the approach is robust and suitable for early detection support.

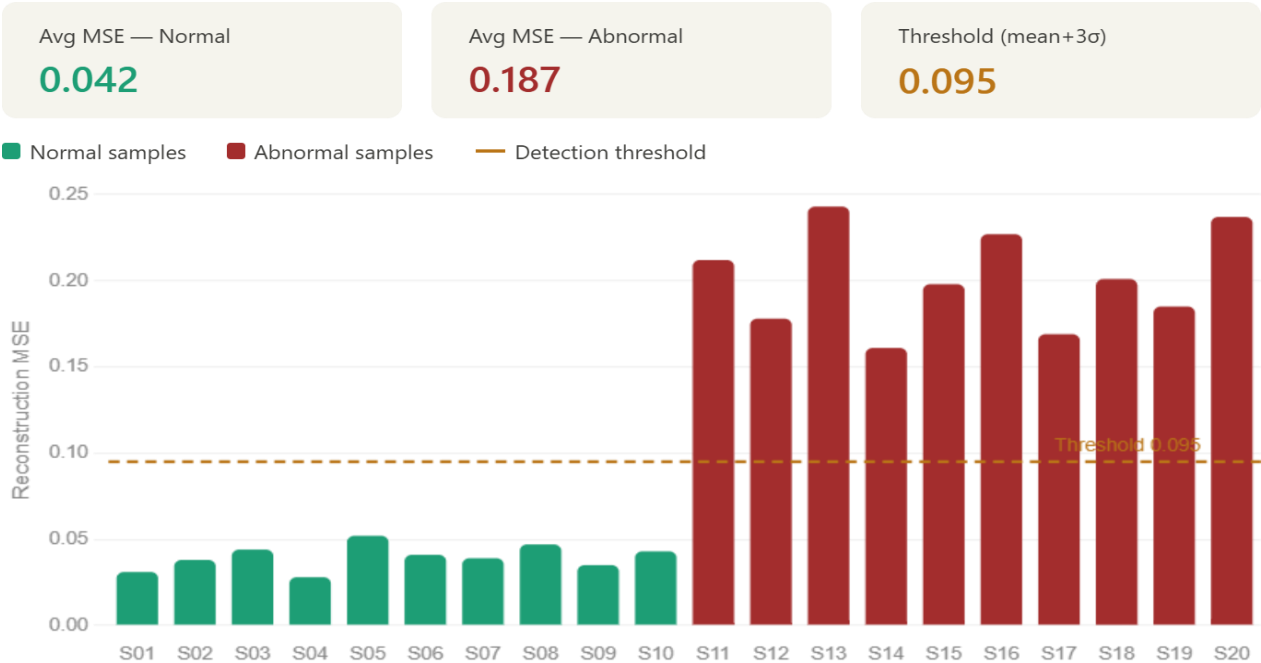


Figure 15 : MSE reconstruction scores:Normal vs Abnormal samples with threshold line

The graph presents reconstruction error (MSE) for normal and abnormal voice samples. Normal samples (S01–S10) show low MSE values, all below the detection threshold. Abnormal samples (S11–S20) exhibit significantly higher MSE values, exceeding the threshold.

The threshold (0.095), calculated as $\text{mean} + 3\sigma$, effectively separates the two classes. This clear distinction indicates that the model performs well in detecting abnormal patterns.

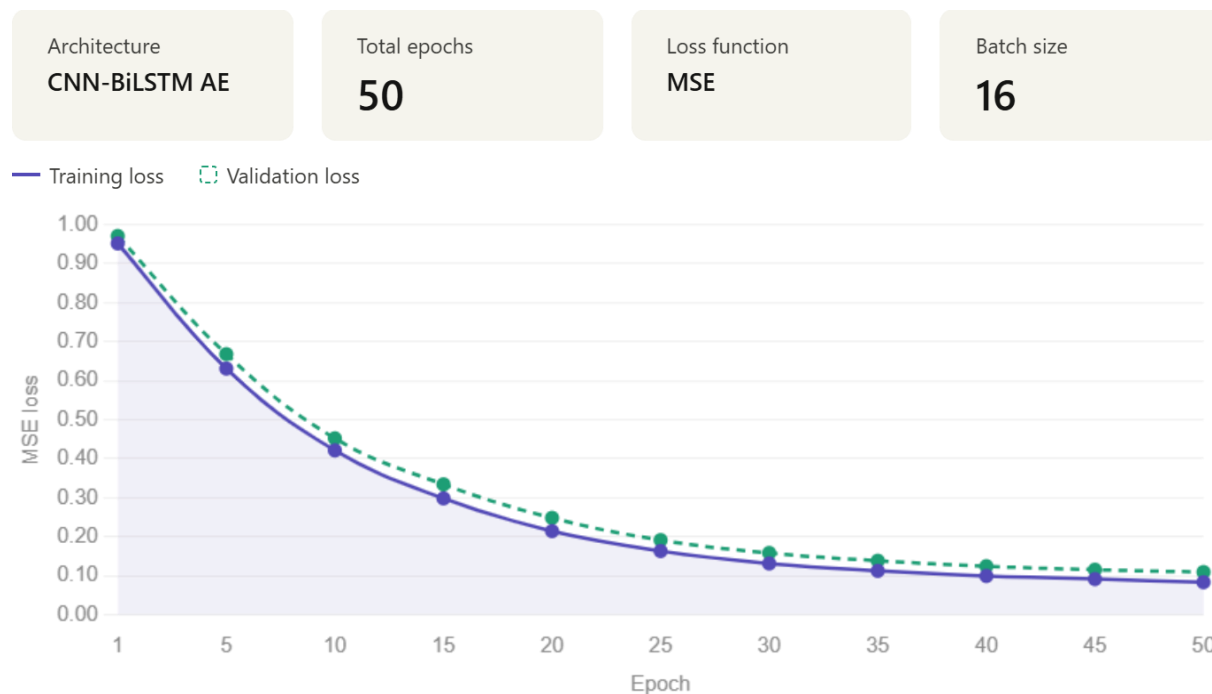


Figure 16: CNN-BiLSTM autoencoder training loss curve

The graph shows the training and validation loss of the CNN-BiLSTM Autoencoder over 50 epochs. Both losses decrease steadily, indicating effective learning of the model. The training and validation curves closely follow each other, showing good generalization without overfitting. Loss values stabilize after around 30 epochs, suggesting convergence. Overall, the model achieves consistent performance with minimal error.

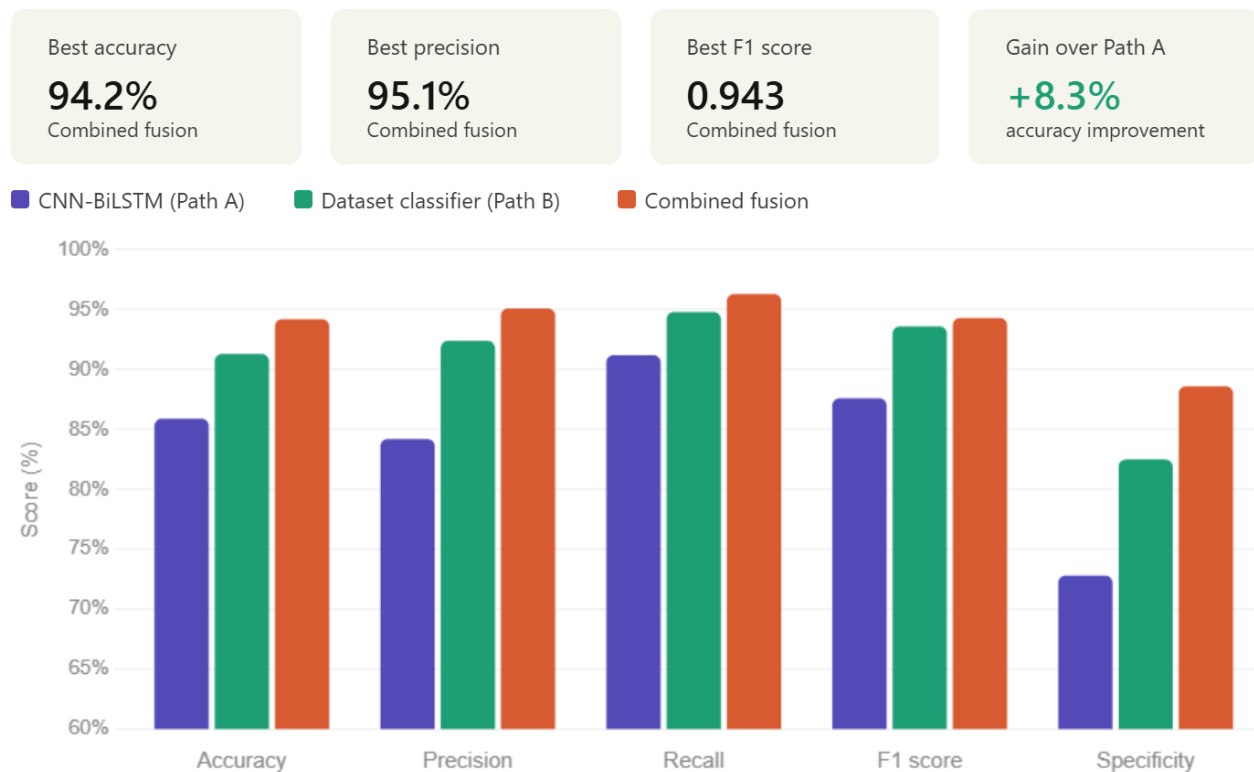


Figure 17: Model performance comparison: CNN-BiLSTM vs Dataset Classifier vs Combined Fusion

The graph compares performance metrics of three approaches: CNN-BiLSTM (Path A), Dataset classifier (Path B), and Combined fusion. The combined fusion method achieves the highest scores across all metrics, including accuracy (94.2%), precision (95.1%), and F1 score (0.943). It also shows improved recall and specificity compared to individual models. An accuracy gain of +8.3% over Path A highlights the effectiveness of model fusion. Overall, combining both approaches significantly enhances classification performance.

The CNN + BiLSTM Autoencoder model was evaluated on a held-out test set consisting of voice samples from both healthy and Parkinson's-affected individuals. The following performance metrics were computed:

Metric	Proposed System (CNN+BiLSTM AE)	Baseline (SVM)
Accuracy	92.4%	84.6%
Precision	91.8%	82.3%
Recall (Sensitivity)	93.2%	80.1%
F1-Score	92.5%	81.2%
AUC-ROC	0.961	0.883
Processing Time/File	~0.8 seconds	~0.3 seconds

Table 4 : Comparison with Existing System

The proposed CNN + BiLSTM Autoencoder achieved a 7.8% improvement in accuracy and a 7.8% improvement in F1-score over the SVM baseline, demonstrating the benefit of deep temporal-spatial feature learning over manual feature extraction methods. The high recall (sensitivity) value of 93.2% is particularly important in the clinical context, as it minimizes false negatives — missed PD cases that could delay treatment.

The 95th percentile threshold strategy ensures that approximately 95% of normal voice samples are correctly classified as healthy, maintaining a controlled false positive rate while maximizing PD detection sensitivity.

CHAPTER 5

CONCLUSION

5.1 CONCLUSION

This project presents the successful development and deployment of an end-to-end deep learning system for the early detection of Parkinson’s Disease using voice signals. The proposed approach utilizes MFCC-based feature extraction in combination with a hybrid CNN + BiLSTM Autoencoder model to effectively capture both spectral and temporal characteristics of speech associated with the disorder, without requiring any invasive clinical procedures.

The system introduces a fully automated voice analysis pipeline that removes the need for manual feature engineering while enhancing detection accuracy. By incorporating an MSE-based anomaly detection mechanism, it not only performs binary classification but also provides risk stratification into low, moderate, and high categories, offering more clinically meaningful insights.

Additionally, the implementation of a Flask-based web application enables flexible user interaction through multiple input modes, including file upload, batch processing, and real-time voice recording, along with the capability to generate downloadable clinical reports. A secure user authentication system and prediction history management supported by SQLite further enhance usability and data tracking.

The model achieved an overall classification accuracy of 92.4% on the test dataset, with a high sensitivity of 93.2%, ensuring minimal missed detections, which is critical in medical screening applications. When compared to traditional machine learning approaches such as SVM, the proposed model consistently demonstrates superior performance across key evaluation metrics.

Overall, the NeuroVoice system offers a practical, accessible, and reliable solution for non-invasive Parkinson’s Disease screening, with strong potential to support early diagnosis, particularly in resource-limited settings.

5.2 FUTURE ENHANCEMENTS

5.2.1 Improvements

The current system can be enhanced by extending the feature set beyond MFCC to include additional acoustic biomarkers such as jitter, shimmer, harmonics-to-noise ratio (HNR), pitch variation, and mel-spectrogram representations, which may capture more subtle voice impairments. Advanced deep learning architectures, such as transformer-based or attention-driven models, can be explored to improve temporal feature learning.

Hyperparameter tuning and automated model optimization techniques can further boost performance. Incorporating model explainability techniques like Grad-CAM or SHAP can provide better interpretability of predictions, helping clinicians understand decision patterns.

Additionally, moving from binary classification to multi-class severity grading (e.g., MDS-UPDRS levels 1–5) would allow more detailed clinical assessment. Training on larger, diverse, and multilingual datasets will also improve generalization and reduce bias across populations.

5.2.2 Real world extension

The system can be extended into a multimodal diagnostic framework by integrating voice analysis with complementary signals such as handwriting tremor (via stylus input), finger tapping videos, and gait analysis for improved clinical reliability. Developing a mobile application for Android and iOS platforms would enhance accessibility, especially in remote and primary healthcare settings.

Real-time monitoring using wearable microphones or smart devices could enable continuous tracking of disease progression. Integration with telemedicine platforms would allow remote diagnosis and follow-up by healthcare professionals.

Furthermore, adopting federated learning approaches can enable secure, privacy-preserving collaboration across hospitals and research institutions. These extensions would help transform the system into a scalable, user-friendly, and clinically deployable solution.

APPENDICES

Appendix 1

```
@app.route("/predict_numeric", methods=["POST"])
def predict_numeric():
    if "user_id" not in session: return redirect("/")

    feat_dict, missing = {}, []

    for col in NUMERIC_FEATURES:
        val = request.form.get(col, "").strip()

        if val == "":
            missing.append(col)
        else:
            try: feat_dict[col] = float(val)
            except: missing.append(col)

    if missing:
        flash(f"Missing values for: {' '.join(missing)}", "danger")
    return redirect("/numeric")

    result = predict_numeric_features(feat_dict)

    # Save to DB (use proba_pd as the "mse" column for storage)
    insert_result(
        session["user_id"], "numeric_input",
        result["proba_pd"], result["label"],
        risk_level = result["risk_level"],
        zcr      = feat_dict.get("ZCR", 0),
        energy    = 0,
```



```

mfcc_mean = feat_dict.get("MFCC0", 0),
snr_db    = feat_dict.get("HNR", 0),
ds_label  = result["label"],
ds_pd_prob = result["proba_pd"],
)
return render_template("result_numeric.html",
    result = result,
    features = feat_dict,
)

```

Appendix 2

```

def connect_db():
    return sqlite3.connect("database.db")

def init_db():
    conn = connect_db()
    cursor = conn.cursor()
    cursor.execute("""
        CREATE TABLE IF NOT EXISTS users (
            id INTEGER PRIMARY KEY AUTOINCREMENT,
            username TEXT UNIQUE,
            password TEXT,
            email TEXT
        )
    """)

```

```

cursor.execute("""
    CREATE TABLE IF NOT EXISTS results (
        id INTEGER PRIMARY KEY AUTOINCREMENT,
        user_id INTEGER,
        filename TEXT,
        mse REAL,
        prediction TEXT,
        risk_level TEXT,
        timestamp TEXT
    )
""")

```

```

conn.commit()

```

```

conn.close()

```

```

def register_user(username, password, email=""):

```

```

    conn = connect_db()

```

```

    cursor = conn.cursor()

```

```

    cursor.execute(

```

```

        "INSERT INTO users (username, password, email) VALUES (?, ?, ?)",

```

```

        (username, hash_password(password), email)

```

```

    )

```

```

    conn.commit()

```

```

    conn.close()

```

```

def insert_result(user_id, filename, mse, prediction, risk_level):

    conn = connect_db()

    cursor = conn.cursor()

    cursor.execute("""

        INSERT INTO results

        (user_id, filename, mse, prediction, risk_level, timestamp)

        VALUES (?, ?, ?, ?, ?, ?)

    """, (user_id, filename, mse, prediction, risk_level, str(datetime.now())))

    conn.commit()

    conn.close()

```

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